

International Quantum Computing Driving License

IQCDL Foundation Level

رخصة القيادة الدولية للحوسبة الكمية - المستوى التأسيسي



✓ Quantum for Everyone

✓ الحوسبة الكمية للجميع

📅 3-Day Program | برنامج 3 أيام

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Course Overview | نظرة عامة على الدورة

Foundation Level - 3-Day Comprehensive Program



Program Structure

Day 1: Quantum Fundamentals

Introduction to quantum computing, qubits, superposition, entanglement, and no-code quantum tools (IBM Composer, Quirk).

أساسيات الحوسبة الكمية والأدوات بدون برمجة

Day 2: Cryptography & Quantum Threat

Current cryptography (RSA, ECC, TLS), quantum threat (Shor's & Grover's algorithms), and Post-Quantum Cryptography (PQC).

التشفير الحالي والتهديد الكمي وتشفير ما بعد الكم

Day 3: Migration & Readiness

EU roadmap, organizational readiness (Assess-Plan-Implement), sector-specific use cases, and call to action.

خارطة طريق الاتحاد الأوروبي والجاهزية المؤسسية



Learning Objectives



Understand quantum computing principles and differentiate from classical computing



Recognize the quantum threat to current cryptography and the urgency of PQC migration



Use no-code quantum tools to build and visualize quantum circuits



Apply Mosca's Theorem to assess organizational quantum risk



Identify sector-specific quantum readiness strategies

Certification Requirements

150 Multiple Choice Questions | 70% Pass Mark (105 correct) | 90 Minutes

سؤال اختيار متعدد | 70% للنجاح | 90 دقيقة 150

Introduction to Quantum Computing

مقدمة في الحوسبة الكمية

Quantum computing represents a fundamental shift in how we process information. Unlike classical computers that use bits (0 or 1), quantum computers leverage the principles of quantum mechanics—superposition, entanglement, and interference—to perform computations that are impossible for classical systems.

تمثل الحوسبة الكمية تحولاً جذرياً في كيفية معالجة المعلومات، مستفيدة من مبادئ ميكانيكا الكم لإجراء حسابات مستحيلة على الأنظمة الكلاسيكية.



Qubits

الكيوبتات



Superposition

التراكب



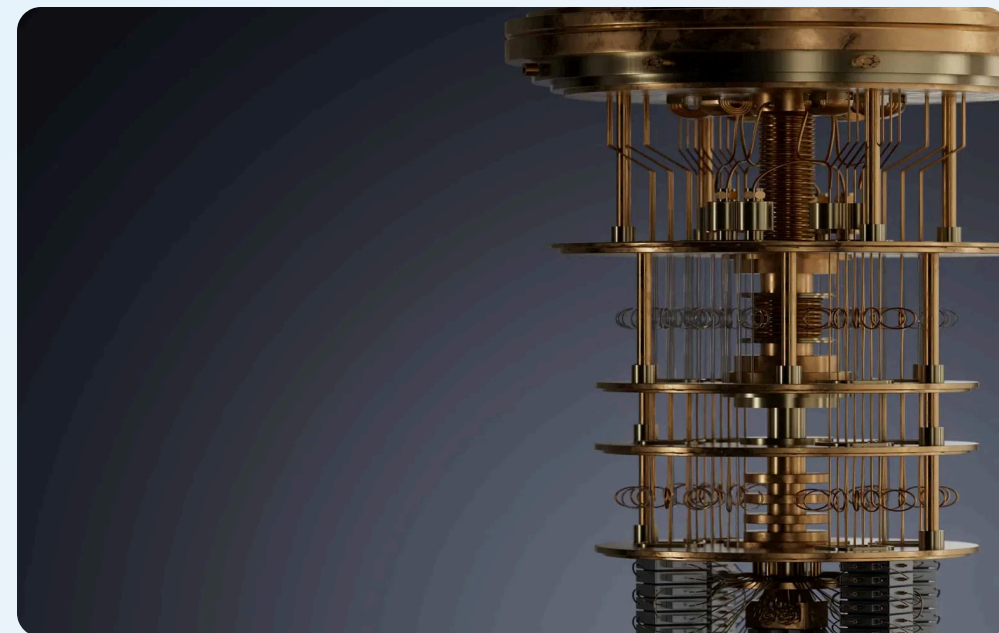
Entanglement

التشابك



Interference

التداخل



Understanding Qubits | فهم الكيوبتات

The Fundamental Unit of Quantum Information

Property	Classical Bit	Quantum Qubit
States	0 or 1	$ 0\rangle$, $ 1\rangle$, or superposition
Representation	Binary digit	Complex vector
Measurement	Direct reading	Collapses to $ 0\rangle$ or $ 1\rangle$
Operations	Logic gates	Quantum gates
Information	1 bit per bit	Exponential capacity

💡 Superposition Principle

A qubit can exist in a superposition of both $|0\rangle$ and $|1\rangle$ states simultaneously, represented as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $|\alpha|^2 + |\beta|^2 = 1$. This allows quantum computers to explore multiple computational paths in parallel.

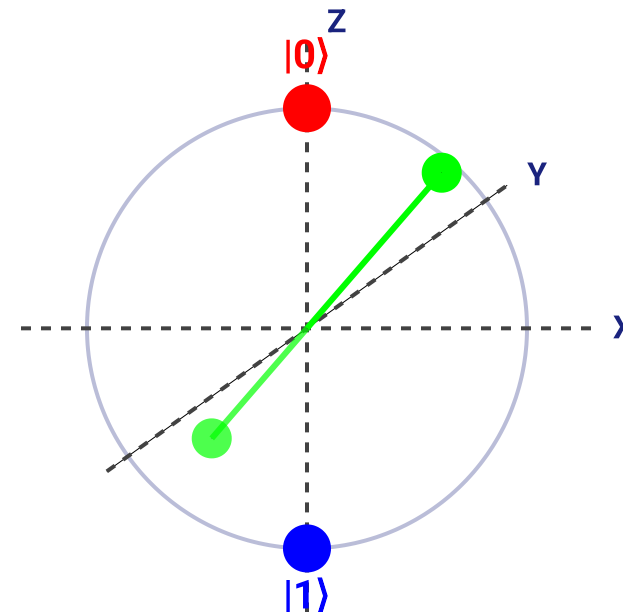
يمكن للكيوبت أن يكون في حالة تراكب لكل من $|0\rangle$ و $|1\rangle$ في نفس الوقت

⚠️ Measurement Collapse

When a qubit is measured, it collapses from superposition to a definite state ($|0\rangle$ or $|1\rangle$) with probabilities $|\alpha|^2$ and $|\beta|^2$ respectively. This irreversible process is fundamental to quantum mechanics.

Bloch Sphere

Qubit Visualization - كرة بلوخ



- $|0\rangle$ State - North pole
- $|1\rangle$ State - South pole
- Superposition - Surface point

Quantum Computing Applications

تطبيقات الحوسبة الكمية في مختلف القطاعات



Drug Discovery

اكتشاف الأدوية

Quantum computers can simulate molecular interactions at the quantum level, dramatically accelerating the discovery of new pharmaceuticals and materials.

- Molecular Simulation
- Protein Folding
- Drug Design



Financial Modeling

النمذجة المالية

Quantum algorithms can optimize portfolios, price complex derivatives, and assess risk with unprecedented speed and accuracy.

- Portfolio Optimization
- Risk Analysis
- Fraud Detection



Optimization Problems

مشاكل التحسين

Quantum computers excel at solving complex optimization problems such as supply chain logistics, traffic routing, and resource allocation.

- Supply Chain
- Traffic Flow
- Scheduling



AI & Machine Learning

الذكاء الاصطناعي والتعلم الآلي

Quantum machine learning algorithms can process high-dimensional data and create complex feature spaces, potentially accelerating AI development.

- Pattern Recognition
- Classification
- Data Analysis



Climate Modeling

نمذجة المناخ

Quantum simulations can model complex climate systems with greater accuracy, helping scientists understand climate change.

- Weather Prediction
- Climate Analysis
- Environmental Science



Cryptography

التشفير

While quantum computers threaten current encryption, they also enable quantum key distribution (QKD) and drive post-quantum cryptography development.

- Quantum Key Distribution
- Post-Quantum Crypto
- Secure Communications



Quantum Technology Market: \$106 Billion by 2040

من المتوقع أن يصل سوق التكنولوجيا الكمية إلى 106 مليار دولار بحلول عام 2040

No-Code Quantum Tools

أدوات الحوسبة الكمية بدون برمجة



IBM Quantum Composer

IBM Quantum Experience

A visual, drag-and-drop quantum circuit builder that allows you to create, simulate, and run quantum algorithms on real quantum computers without writing any code.

أداة مرئية لبناء الدوائر الكمية بالسحب والإفلات

- ✓ Visual circuit design with drag-and-drop gates
- ✓ Access to real IBM quantum hardware
- ✓ Built-in simulators for testing
- ✓ Bloch sphere and statevector visualization



Quirk Quantum Simulator

Open Source Browser Tool

A browser-based quantum circuit simulator that provides real-time visualization of quantum states as you build circuits. Perfect for learning and experimentation.

محاكي دوائر كمية قائم على المتصفح

- ✓ Instant real-time visualization
- ✓ No installation or account required
- ✓ Interactive probability displays
- ✓ Educational and intuitive interface

Common Quantum Gates | البوابات الكمية الشائعة

H

Hadamard

Creates superposition

يخلق التراكب

X

Pauli-X (NOT)

Bit flip

قلب البت

CNOT

Controlled-NOT

Creates entanglement

يخلق التشابك

Z

Pauli-Z

Phase flip

قلب الطور

M

Measurement

Reads qubit state

قراءة الكيوبت



Getting Started | البدء

Access IBM Quantum Composer with a free IBM Quantum account

quantum-computing.ibm.com

Current Cryptography & The Quantum Threat

التشفير الحالي والتهديد الكمي

🔒 Current Cryptography

🔑 Symmetric Encryption

AES - Uses a single shared key for encryption and decryption. Fast and efficient for bulk data.

التشفير المتماثل - مفتاح واحد مشترك

↔ Asymmetric Encryption

RSA & ECC - Public-private key pairs for secure key exchange and digital signatures.

التشفير غير المتماثل - زوج من المفاتيح

🌐 TLS/SSL Protocols

Transport Layer Security - Secures HTTPS connections, protecting billions of devices worldwide.

تؤمن اتصالات الإنترنت - TLS بروتوكولات

☢️ The Quantum Threat

📊 Shor's Algorithm

Breaks RSA and ECC by efficiently factoring large numbers. Exponential speedup over classical methods.

ECC و RSA خوارزمية شور - تكسر

🔍 Grover's Algorithm

Provides quadratic speedup for searching. Reduces effective key length of symmetric encryption.

خوارزمية جروفر - تضعف التشفير المتماثل

📦 Harvest Now, Decrypt Later

Adversaries collect encrypted data today to decrypt when quantum computers become available.

اجمع الآن، فك التشفير لاحقاً - تهديد حالي

⚠️ 60+ Billion Devices

Rely on RSA/ECC vulnerable to quantum attacks

المعرضة للهجمات الكمية RSA/ECC تعتمد على

🕒 Timeline: 2030-2040

Estimated arrival of CRQC (Cryptographically Relevant Quantum Computer)

الوقت المقدر لوصول الحاسوب الكمي المؤثر تشفيراً

التهديد الكمي | The Quantum Threat

Breaking Today's Cryptography

⚠️ Shor's Algorithm

Developed by Peter Shor in 1994, this quantum algorithm can factor large numbers exponentially faster than classical algorithms. It breaks RSA and ECC, the foundation of public-key cryptography used by billions of devices worldwide.

أساس التشفير بالمفتاح العام - RSA و ECC خوارزمية شور تكسر

🔍 Grover's Algorithm

Provides quadratic speedup for searching. Reduces effective key length of symmetric encryption (like AES) by half. AES-128 becomes as secure as AES-64, requiring migration to AES-256.

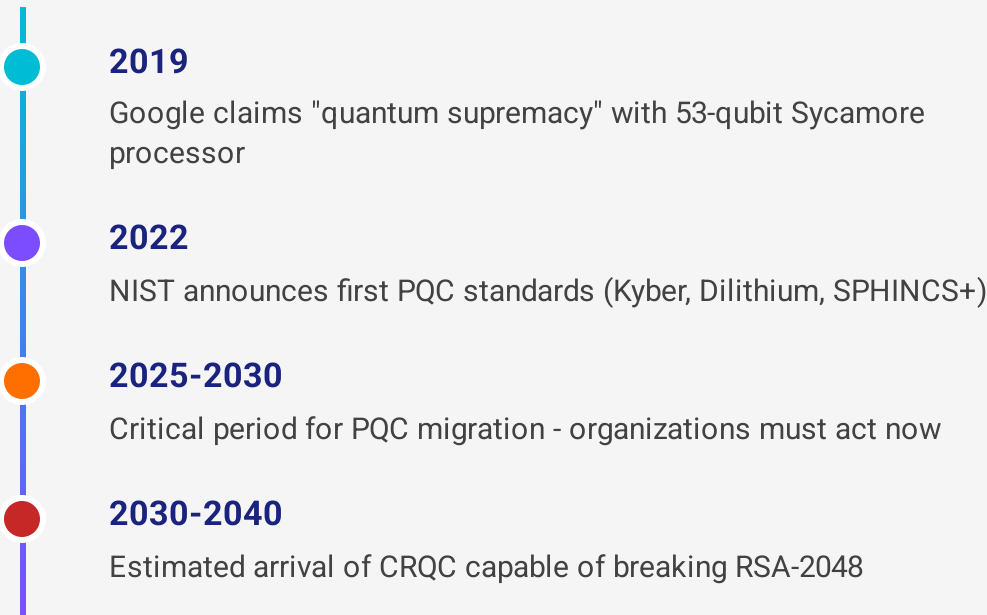
خوارزمية جروفر تقلل فعالية مفاتيح التشفير المتماثل

📦 "Harvest Now, Decrypt Later"

Adversaries are collecting encrypted data today with the intent to decrypt it once a CRQC becomes available. Data with long-term confidentiality requirements is at risk NOW.

البيانات المشفرة اليوم معرضة للخطر - التهديد حقيقي الآن

🕒 Timeline to CRQC



Vulnerable Systems

- ✗ RSA-2048 & RSA-4096 encryption
- ✗ Elliptic Curve Cryptography (ECC)
- ✗ Diffie-Hellman key exchange
- ✗ Digital signatures (RSA, ECDSA)
- ✗ TLS/SSL certificates
- ✗ VPN connections (IPsec)

Mosca's Theorem: $X + Y > Z$

نظرية موسكا - إطار تقييم المخاطر الكمية

$$X + Y > Z$$

When this inequality is true, your data is at risk

عندما تكون هذه المعادلة صحيحة، فإن بياناتك معرضة للخطر

X

Data Security Shelf-Life

عمر أمان البيانات

How long must your data remain confidential?
Government: 50+ years. Healthcare: 30-50 years.
Financial: 7-10 years.

Y

Migration Time

وقت الانتقال

How long to migrate to PQC? Large enterprises: 5-10 years. Critical infrastructure: 10-15 years. Complete ecosystem: 15-20 years.

Z

Time to CRQC

CRQC الوقت حتى

When will CRQC exist? Conservative: 2030-2040. Optimistic: 2028-2035. Must plan for earliest scenario.



Safe (For Now)

$$X + Y < Z$$



At Risk - Act Now!

$$X + Y > Z$$

Post-Quantum Cryptography (PQC)

تشفير ما بعد الكم

Post-Quantum Cryptography refers to cryptographic algorithms designed to be secure against attacks by both classical and quantum computers. In 2022, NIST standardized the first set of PQC algorithms to replace vulnerable RSA and ECC systems.

تشفير ما بعد الكم - خوارزميات آمنة ضد الحواسيب الكلاسيكية والكمية



Key Encapsulation

تغليف المفاتيح

CRYSTALS-Kyber

قائم على الشبكات | Lattice-Based

Primary algorithm for key establishment. Efficient performance with relatively small key sizes. Based on Module Learning With Errors (MLWE) problem.



Digital Signatures

التوقيعات الرقمية

CRYSTALS-Dilithium

قائم على الشبكات | Lattice-Based

Primary signature algorithm. Balances security, signature size, and speed. Based on MLWE and MSIS.

SPHINCS+

قائم على التجزئة | Hash-Based

Stateless hash-based signature. Conservative security based on well-understood hash functions.



✓ NIST Standardization Complete (2022-2024)

National Institute of Standards and Technology | المعهد الوطني للمعايير والتكنولوجيا

FIPS 203, 204, 205

Federal Information Processing Standards

PQC Migration & Organizational Readiness

الانتقال إلى تشفير ما بعد الكم والجاهزية المؤسسية

📍 EU PQC Roadmap

- 2026

National Strategies Ready
Member states develop PQC transition strategies and begin inventory phase.
- 2030

Critical Infrastructure Protected
Essential services, government, and critical sectors complete PQC migration.
- 2035

Full Sector Transition
All sectors complete migration. Classical algorithms decommissioned.

☰ Three-Phase Approach

- 1

Assess | التقييم
Inventory all cryptographic assets (CBOM), identify vulnerabilities, prioritize systems.
- 2

Plan | التخطيط
Develop migration roadmap, test PQC algorithms, design hybrid solutions.
- 3

Implement | التنفيذ
Deploy PQC, update systems, train teams, monitor and validate security.

🔄 Crypto-Agility

The ability to quickly switch cryptographic algorithms without disrupting operations. Essential for responding to new threats and algorithm updates.

المرونة التشفيرية - القدرة على تبديل الخوارزميات بسرعة

- ✓

Abstract crypto functions
- ✓

Modular architecture
- ✓

Centralized key management
- ✓

Automated updates

📊 Mosca's Theorem

$$X + Y > Z$$

X: Data security shelf-life (how long data must stay confidential)

Y: Migration time (time needed to transition to PQC)

Z: Time until CRQC is available

If $X + Y > Z$, your data is at risk!

إفبياناتك في خطر، إذا كان $X + Y > Z$

⚠️ Act Now, Not Later

Organizations must begin PQC migration today to protect tomorrow's data
يجب على المؤسسات البدء في الانتقال اليوم لحماية بيانات الغد

Sector-Specific Use Cases

حالات الاستخدام الخاصة بالقطاعات



Telecommunications

الاتصالات

CRITICAL RISK

Telecom infrastructure relies heavily on public-key cryptography for network security, VPNs, and 5G authentication. A quantum attack could compromise entire communication networks.

- ✓ Migrate core network encryption to PQC
- ✓ Update VPN and IPsec implementations
- ✓ Implement hybrid TLS for customer services
- ✓ Coordinate with equipment vendors on PQC support



Banking & Finance

البنوك والمالية

CRITICAL RISK

Financial transactions, digital signatures, and secure communications are vulnerable. Long-term financial records and contracts require immediate protection.

- ✓ Secure payment gateways with PQC algorithms
- ✓ Update digital signature systems (contracts, transactions)
- ✓ Protect long-term financial records and archives
- ✓ Ensure PCI-DSS compliance with quantum-safe standards



Healthcare

الرعاية الصحية

HIGH RISK

Electronic Health Records (EHR) must remain confidential for decades. Patient privacy regulations (HIPAA) require long-term data protection against future threats.

- ✓ Secure EHR systems with PQC encryption
- ✓ Protect medical device communications
- ✓ Update telemedicine platforms for quantum safety
- ✓ Ensure HIPAA compliance with PQC standards



Retail & E-Commerce

التجزئة والتجارة الإلكترونية

MEDIUM RISK

Online payment systems, customer data, and supply chain communications need quantum-safe protection to maintain trust and regulatory compliance.

- ✓ Upgrade e-commerce platforms to hybrid TLS
- ✓ Secure payment processing with PQC
- ✓ Protect customer databases and loyalty programs
- ✓ Coordinate with payment processors on PQC adoption

🚩 Prioritization: Critical Infrastructure First

Focus on systems with long-lived data, high-value assets, and difficult-to-update infrastructure

ركز على الأنظمة ذات البيانات طويلة الأمد والأصول عالية القيمة والبنية التحتية الصعبة التحديث

IQCDL Certification Path

مسار شهادة رخصة القيادة الدولية للحوسبة الكمية

LEVEL 1



Foundation Level

المستوى التأسيسي

🕒 3 Days Training

- ✓ Quantum computing fundamentals
- ✓ Post-quantum cryptography basics
- ✓ Organizational readiness strategies
- ✓ No-code tools and simulations
- ✓ 150 MCQ exam



LEVEL 2



Practitioner Level

مستوى الممارس

🕒 5 Days Training

- ✓ Hands-on quantum programming
- ✓ PQC algorithm implementation
- ✓ Real-world case studies
- ✓ Migration project planning
- ✓ 150 MCQ exam



Foundation Exam

امتحان المستوى التأسيسي

150

Questions

سؤال

180

Minutes

دقيقة

70%

Pass Score

MCQ

Format



Practitioner Exam

امتحان مستوى الممارس

150

Questions

سؤال

240

Minutes

دقيقة

75%

Pass Score

MCQ

Format

دعوة للعمل | Call to Action

The Time to Act is NOW



Why Act Now?

لماذا نتصرف الآن؟

- 1 Migration Takes Time:** Large organizations need 5-10 years to fully transition to PQC. Starting today means you'll be ready before CRQC arrives.
- 2 Harvest Now, Decrypt Later:** Your encrypted data is being collected today. Waiting means your current data will be vulnerable retroactively.
- 3 Competitive Advantage:** Early adopters will gain trust, compliance, and market leadership while others scramble to catch up.
- 4 Regulatory Pressure:** Governments and industry bodies are mandating PQC adoption. Compliance deadlines are approaching fast.



Who Should Lead?

من يجب أن يقود؟

- CISOs & Security Leaders:** Champion PQC migration and build organizational awareness
- IT & DevOps Teams:** Implement crypto-agility and test PQC algorithms
- Business Leaders:** Allocate budget and prioritize quantum readiness initiatives
- Compliance Officers:** Ensure alignment with emerging PQC regulations
- Everyone:** Build quantum literacy and advocate for organizational readiness

Your Next Steps

خطواتك التالية



STEP 1

Assess: Inventory your cryptographic assets and identify vulnerabilities



STEP 2

Learn: Continue to Practitioner Level for hands-on PQC implementation



STEP 3

Act: Begin your organization's PQC migration journey today

⚠ The Quantum Threat is Real. The Time to Prepare is NOW.